Date of publication xxxx 00, 0000, date of current version xxxx 00, 0000.

Digital Object Identifier 10.1109/ACCESS.2025.DOI

Link Prediction via Adaptive Teacher–Student Graph Neural Networks for Drug Repurposing on Heterogeneous Biomedical Knowledge Graphs

PRAKHAR[1], ARYAN[1], and ADITYA[1]

[1]School of Computer Engineering, Manipal Institute of Technology, Manipal Academy of Higher Education, Manipal 576104, India (e-mail: prakhar2.mitmpl2023@learner.manipal.edu; aryan9.mitmpl2023@learner.manipal.edu; aditya24.mitmpl2023@learner.manipal.edu)

Corresponding author: Prakhar (e-mail: prakhar2.mitmpl2023@learner.manipal.edu).

This work was supported by Manipal Institute of Technology, Manipal Academy of Higher Education, Manipal, India.

ABSTRACT We present an end-to-end drug repurposing pipeline that makes use of link prediction from heterogeneous biomedical knowledge graphs. Using a knowledge distillation framework, we train a compact one-layer Student graph neural network inspired by the two-layer Teacher model.Graphs were created with data from nine different biomedical sources using uncertainty-weighted alignment between such datasets in order to alleviate class imbalance and identify possible drugs based on existing biological interactions. Rich features from the nodes were extracted by combining graph topology, molecular based domain-specific features, and BERT clusters. Within a GDRnet GKDRMTL teacher-student architecture, we implement temperature-annealed KL (Kullback-Leibler) distillation, hidden-state MSE (mean squared error) alignment, and homoscedastic uncertainty weighting as part of our overall efforts to establish the predictive equivalency of both models. The results show that both Student and Teacher models have good predictive accuracy (i.e. Teacher AUROC = 0.976 and AUPRC = 0.947; and Student AUROC = 0.936 and AUPRC = 0.849). Additionally, although the Student model applied with slightly less parameters produces nearly the same results as the Teacher, there was 19 AUROC points of increase between the two when baseline predictions were compared to predictions made with graph-based message passing. We also show that the top predicted drug – disease combinations had intermediate gene-based mechanistic evidence of their underlying biology, and provide a strong biological rationale and starting point for further investigation of these hypotheses in wet lab settings.

INDEX TERMS Biomedical knowledge graphs, drug repurposing, graph neural networks, GraphSAGE, heterogeneous networks, knowledge distillation, link prediction, teacher-student learning.

I. INTRODUCTION

It takes approximately 10 years and more than $2.6 billion on average to develop a new medicine from the time of identifying a disease target to gaining approval from the FDA (U.S. Food and Drug Administration) or another regulatory agency. More than 90% of drugs that enter clinical trials do not make it to approval. Repurposing of existing drugs can redefine how quickly we use medications approved or investigating the treatment of an existing condition but now with a revised indication (drugs can be repurposed into a new indication) and reduce the costs of developing drugs by using the existing safety and manufacturing systems and processes related to that existing drug (for example, sildenafil was originally developed as an anti-anginal but was repurposed for the treatment of pulmonary arterial hypertension). Previously, the discovery of drugs that were successfully repurposed has generally been made coincidentally. In recent years, however, it has become possible to apply computational tools specifically designed to systematically identify and discover repurposing opportunities at scale. [1]–[3].

The use of knowledge graphs (KGs) to represent the complex multirelational networks of drugs, diseases, genes/proteins, and other biological entities, etc., has created a unique opportunity to repurpose drugs. Large resource

collections (e.g., Hetionet, Drug Repurposing Knowledge Graph. GNBR) contain large amounts of heterogeneous datasets of drug and disease pairings that have been consolidated from various curated data sources into a single unified graph structure that is scalable to machine learning (ML) methodologies. Using existing KGs for repurposing provides the ability to identify new drugs that might work for a condition based on probability using the existing and new drug-disease relationships in the KGs. Both traditional methods and graph-based methods of ML will benefit from the structured representation of the drug repurposing process using KGs. [4]–[8].

Link prediction is provided by knowledge graphs (KG) through the use of graph neural networks (GNN), which have become the preferred method. The GDRnet architecture used a SIGN-like architecture to encode multi-hop neighbouring aggregations that were pre-computed and feed into a bilinear decoder, and is able to achieve a high AUROC on biomedical heterogeneous KGs. In parallel, the GKDRMTL framework developed a graph-based knowledge distillation approach using multi-task learning for drug repurposing. With this approach, a compact student model can match or exceed the performance of a larger teacher on medicines that were not included in the drug of choice. Other methods employed for this drug repurposing include DRAGON and graph hierarchical methods. In addition to an emphasis on accuracy, explainability is also becoming more important . This is defined by determining which biological medium (genes, pathways, etc.) acted as mediators for the predicted relationship between a drug and a disease. [7], [9]–[14].

Even though all these advancements have been made to use GNN's for drug repurposing in the real-world, the use of GNN's for this purpose will face three major challenges in their implementation: (1) Very few known drug-disease relationships have been identified (less than 0.1% out of the possible pairings). (2) The various node features are very high dimensional and heterogeneous and consist of many different pieces of topological data (molecular structure, etc.) along with semantic (BERT encoding) information; and (3) To evaluate millions of candidates in at least once time, a GNN-based system must use significant resources to carry out the calculations needed to obtain the results

We created an integrated process to solve all three of these problems (all three of the different problems you listed are the same). The main parts of our work are:

1) A pipeline that constructs a multi-source heterogeneous KG by merging nine different but complementary biomedical datasets and applying BERT-based disease grouping for semantic enrichment.
2) A rich scheme for creating node features by combining graph-topology features, domain-specific molecular features, and BERT cluster embeddings.
3) A Teacher-Student GNN architecture based on the GDRnet & GKDRMTL models using temperature-annealed KL distillation, hidden-state MSE alignment, and homoscedastic uncertainty weighting.
4) Thorough ablation studies of these contributions, evaluation metrics, and a method of interpreting the mechanisms through the identification of intermediate gene nodes.

The contribution builds upon existing GNN approaches for drug repurposing found in prior works on self-supervised and graph-based paradigms. [7], [13], [15], [16].

II. RELATED WORK

A. GRAPH NEURAL NETWORKS FOR DRUG DISCOVERY

GNNs model molecular structures and interaction networks as graphs, allowing for the end-to-end learning of structural and relational patterns. In a survey by Wang et al., the authors provide an exhaustive taxonomy of the many architectures of GNNs used in drug discovery, specifically various components within the discovery pipeline (molecular property predictions to modelling target–ligand interactions). A seminal paper by Zitnik et al. used graph convolutional networks (GCNs) to model polypharmacy (multi-drug use) side effects and laid the groundwork for multi-relational biomedical link predictions. KGNNs (knowledge graph neural networks) and subsequent knowledge graph embedding models extended this body of work for drug–drug interaction (DDI) predictions, with DDIPred showing the potential of using GCN-based drug chemical structure embeddings (the nodes of the graphs) for this task. Other aspects of medical drug analysis in the context of GNNs have been explored in IEEE student research, while the classification of drug effects using frequency-based GNN graph traversals continues to demonstrate the ever-increasing range of GNN applications in pharmacology. The use of deep multimodal graphs and structures in drug–target interaction predictions further builds on the existing GNN framework in drug–target interaction predictions (GENNDTI). [6] [20], [21] [18], [19] [9] [10], [17].

B. DRUG REPURPOSING VIA NETWORK MEDICINE

Morselli Gysi and colleagues utilized a network medicine approach to identify repurposing opportunities for drugs targeted against COVID-19. They found that the network proximity between drug targets and disease modules in the human interactome strongly predicted the efficacy of these repurposed drugs. In addition, Zeng and others have reported on the use of deep neural networks to integrate target-specific molecular descriptors with graph-based embeddings in order to generate candidate drugs for both repurposing and de novo design. The study by Stokes and colleagues demonstrated that the use of AI-guided drug discovery was successful in identifying novel antibiotics through in vivo studies, underscoring the translational utility of computational drug screens. Additionally, deep learning-derived repurposing strategies for COVID-19 and Alzheimer's disease have demonstrated the diversity of applications where predicting links between biomedical literature and knowledge graphs results in actionable predictions. Graph-based

knowledge graphs have also shown to provide predictive and interpretable signals for drug discovery and for assessing the risk of opioid use disorder. Others, such as DRIVE, have provided disease visualization and repurposing capabilities at the platform level. Cancer drug response prediction from heterogeneous GNNs and multi-omics data as well as fast adapting GNNs with prior drug response knowledge further demonstrate the translational relevance of these approaches. [22] [23] [24] [25], [26] [28] [27] [29], [30].

C. KNOWLEDGE DISTILLATION IN BIOMEDICAL AI

Knowledge distillation (KD), introduced by Hinton et al., trains a compact Student model to mimic the output distribution of a larger Teacher, preserving predictive performance while reducing inference cost [31]. Gou et al. survey the landscape of KD techniques [32]. In the drug discovery domain, the GKDRMTL framework adapts KD with multi-task learning and graph-structured knowledge transfer, achieving state-of-the-art repurposing performance [13]. KEDRec-LM further extends this paradigm to large language models with explainability [33]. Predicting drug efficacy with molecular embeddings and graph context demonstrates that combined embedding and distillation-inspired strategies continue to advance the field [34].

D. EXPLAINABLE AI FOR DRUG DISCOVERY

Jimenez-Luna et al. review XAI methods for drug discovery, emphasising attribution methods and mechanistic transparency [11]. GNNExplainer provides instance-level subgraph explanations for GNN predictions and has been widely adopted for biological network interpretation [12]. Our pipeline adopts a simpler but interpretable approach—identifying shared gene/protein intermediary nodes between predicted drug–disease pairs—aligning with the intermediate-node analysis reported in GKDRMTL [13]. Graph neural networks with transformer-enhanced embeddings for drug-target-disease association prediction and the DRIVE platform also incorporate interpretability as a design objective [28], [35]. For COVID-19 in particular, graph neural network approaches have been applied directly to drug discovery tasks, illustrating that explainability considerations extend across diverse disease contexts [36].

III. METHODOLOGY

A. PROBLEM FORMULATION

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{T}_V, \mathcal{T}_E)$ be a typed, undirected heterogeneous graph where $\mathcal{V}$ is the node set, $\mathcal{E}$ the edge set, $\mathcal{T}_V = \{drug, disease, gene/protein\}$ the node type set, and $\mathcal{T}_E$ the relation type set. Let $\mathbf{X} \in \mathbb{R}^{N \times d}$ denote the node feature matrix, where $N = |\mathcal{V}|$. Drug repurposing is cast as link prediction: for each unseen drug–disease pair (v_i, v_j) with $\tau(v_i) = drug$ and $\tau(v_j) = disease$, estimate $p(\hat{y}_{ij} = 1 \mid \mathcal{G}, \mathbf{X})$ [6], [7].

B. MULTI-SOURCE KNOWLEDGE GRAPH CONSTRUCTION

Following the multi-source integration strategy of GDRnet and leveraging Hetionet and DRKG as conceptual foundations, we merge four raw KG files (`kg_csv`, `kg_giant_csv`, `kg_grouped_csv`, `kg_raw_csv`) plus additional processed edge/node files (`edges1`, `nodes1`). After deduplication on the (x_id, y_id) pair key, six typed subgraphs are extracted. Inter-type edges of the *treats/palliates* type represent drug–disease edges used as positive labels. Inter-type edges of the *targets* type capture drug–gene/protein interactions, while inter-type edges of the *causes* type represent gene/protein–disease associations. Three intra-type subgraphs complete the construction: drug–drug similarity edges encoding chemical and pharmacological similarity, disease–disease similarity edges encoding phenotypic and co-morbidity overlap, and gene–gene interaction edges encoding protein–protein interactions. BERT-encoded disease cluster embeddings from `kg_grouped_diseases_bert_map_csv` are incorporated as supplementary node features, providing semantic grouping of related disease concepts beyond structural graph proximity [4], [5], [7].

C. DATA PREPROCESSING

Following the data quality guidance of Wang et al. [6], columns with more than 50% missing rate are dropped. For remaining numeric columns with fewer than 5,000 samples, KNN imputation ($k = 5$) is applied; median imputation is used otherwise. Categorical columns use mode imputation. Node type indicators are encoded as one-hot vectors $\mathbf{K} \in \mathbb{R}^{N \times |\mathcal{T}_V|}$, consistent with GKDRMTL's feature matrix $\mathbf{X} = [\mathbf{R} \| \mathbf{K}]$ formulation. Binary categorical features are label-encoded; low-cardinality features (fewer than 10 unique values) receive full one-hot encoding; high-cardinality features use label encoding. Topology features are standardized with zero-mean, unit-variance scaling, while molecular drug/disease features are min-max scaled to $[0, 1]$. The normalised adjacency $\tilde{\mathbf{A}} = \mathbf{D}^{-1/2}\mathbf{A}\mathbf{D}^{-1/2}$ is used for message passing, as in GDRnet. A three-stage feature selection pipeline is applied: near-zero variance removal (threshold $= 0.01$); pairwise correlation filtering ($r > 0.95$, keeping one representative per correlated cluster); and supervised SelectKBest (ANOVA F-statistic) when labels are available [6], [7], [13].

The positive-to-negative ratio in drug–disease link prediction is typically 1:30 or worse. This imbalance is addressed through a combination of strategies: a positive class weight $w^+ = N^-/N^+$ passed to `BCEWithLogitsLoss`; hard negative sampling biased toward high-degree drug and disease nodes (1:3 explicit ratio, further compensated by w^+); and SMOTE-based oversampling on training feature representations for ancillary baselines. Fig. 1 visualises the class distribution across splits, the precision–recall trade-off as a function of positive class weight, and the PCA projection of drug–disease pair features [7].

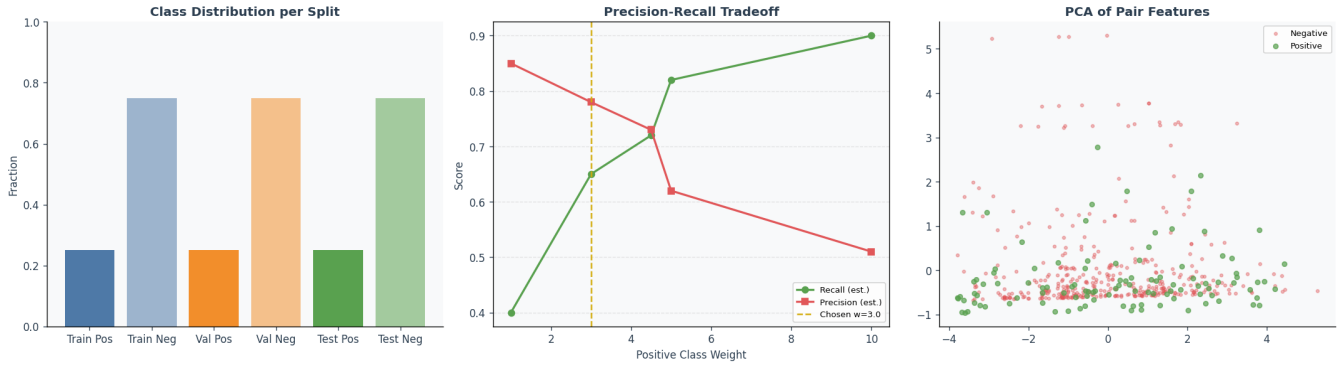

FIGURE 1. Class imbalance analysis. *Left:* Positive/negative fraction across train, validation, and test splits (see also Table 1). *Centre:* Precision–recall trade-off as a function of positive class weight w^+; the dashed line marks the chosen value $w^+ = 3.0$. *Right:* PCA projection of concatenated drug–disease pair features coloured by class label, illustrating the feature-space overlap that motivates GNN-based representation learning.

D. NODE FEATURE ENGINEERING

The final node feature matrix $\mathbf{X} \in \mathbb{R}^{N\times d}$ (where $d = 8 + 10 + d_{\text{BERT}}$) concatenates three feature blocks. The topology feature block (8-dim) consists of normalised degree, local clustering coefficient, approximate betweenness centrality, PageRank score, normalised node-type index, and a 3-dim one-hot node type vector. The domain feature block (10-dim) comprises the top-10 selected columns from the drug or disease feature CSV files, zero-padded for non-matching node types. The BERT cluster feature block (d_{BERT}-dim) provides BERT-based disease grouping embeddings from the semantic disease map, contributing latent semantic similarity among disease concepts.

E. TEACHER–STUDENT GNN ARCHITECTURE

Fig. 2 illustrates the overall Teacher–Student framework. BERT, topology, and domain-specific node features are fed into a 2-layer GraphSAGE Teacher and a 1-layer GraphSAGE Student. Knowledge distillation aligns the two models through hidden-state MSE loss and temperature-annealed KL divergence loss, while a shared bilinear scoring function maps learned drug and disease embeddings to drug–disease association probabilities.

1) Teacher Model

The Teacher is a two-layer GraphSAGE encoder augmented with skip connections, batch normalisation, and an edge-type bias embedding, followed by a learnable bilinear decoder. For layers $l = 1, 2$:

$$\mathbf{h}_v^{(l)} = \text{BN}\left(\text{SAGEConv}^{(l)}\left(\mathbf{h}_v^{(l-1)}, \left\{\mathbf{h}_u^{(l-1)}\right\}_{u\in\mathcal{N}(v)}\right)\right) \tag{1}$$

with LeakyReLU activations and dropout ($p = 0.3$). A residual skip connection $\mathbf{h}_v^{(2)} \mathrel{+}= \mathbf{W}_{\text{skip}}\mathbf{x}_v$ is added, following GDRnet's residual design. The hidden dimension is 256 and the output embedding dimension is 128. Given drug embedding $\mathbf{z}_{c_i}$ and disease embedding $\mathbf{z}_{d_j}$, the link score follows GDRnet Equation 3:

$$s_{ij} = \sigma\left(\mathbf{z}_{c_i}^\top \mathbf{\Phi}\, \mathbf{z}_{d_j}\right) \tag{2}$$

where $\mathbf{\Phi} \in \mathbb{R}^{128\times 128}$ is a learnable interaction matrix, initialised as $\mathbf{I} + 0.01\,\mathcal{N}(0,1)$ [7], [37].

2) Student Model

The Student is a one-layer GraphSAGE encoder with hidden dimension 128 and output dimension 64, plus a projection layer $\mathbf{P} \in \mathbb{R}^{64\times 128}$ for hidden-state alignment. The Student shares the bilinear decoder structure of (2).

3) Knowledge Distillation Loss

Following GKDRMTL [13], the Student training objective combines three terms. The task loss (GDRnet Eq. 4) is weighted binary cross-entropy:

$$\mathcal{L}_{\text{task}} = -\frac{1}{|\mathcal{B}|}\sum_{(i,j)\in\mathcal{B}}\left[w^+ y_{ij}\log\hat{p}_{ij} + (1-y_{ij})\log(1-\hat{p}_{ij})\right] \tag{3}$$

The distillation loss (GKDRMTL Eq. 16) applies temperature-scaled KL divergence:

$$\mathcal{L}_{\text{distill}} = T^2 \cdot \text{KL}\left[\sigma\left(\frac{\hat{s}^T}{T}\right) \,\middle\|\, \sigma\left(\frac{\hat{s}^S}{T}\right)\right] \tag{4}$$

where $T \in [T_{\min}, T_{\max}]$ follows a cosine annealing schedule (GKDRMTL Eq. 17). The hidden-state alignment term (GKDRMTL Eq. 18) computes a projected MSE between Student and Teacher embeddings:

$$\mathcal{L}_{\text{HS}} = \left\|\mathbf{P}\mathbf{H}^S - \mathbf{H}^T\right\|_F^2 \tag{5}$$

The total Student loss with homoscedastic uncertainty weighting is:

$$\mathcal{L} = \alpha\tilde{\mathcal{L}}_{\text{task}} + \beta\tilde{\mathcal{L}}_{\text{distill}} + \gamma\mathcal{L}_{\text{HS}} \tag{6}$$

where $\alpha = 0.5$, $\beta = 0.3$, $\gamma = 0.2$, and $\tilde{\mathcal{L}}_k = e^{-\sigma_k}\mathcal{L}_k + 0.5\sigma_k$ for learnable log-variance σ_k per task (GKDRMTL Eq. 19–20) [13], [38].

F. TRAINING PROTOCOL

The Teacher is trained for 100 epochs using the Adam optimiser (lr $= 8\times 10^{-4}$, weight decay $= 10^{-4}$) with

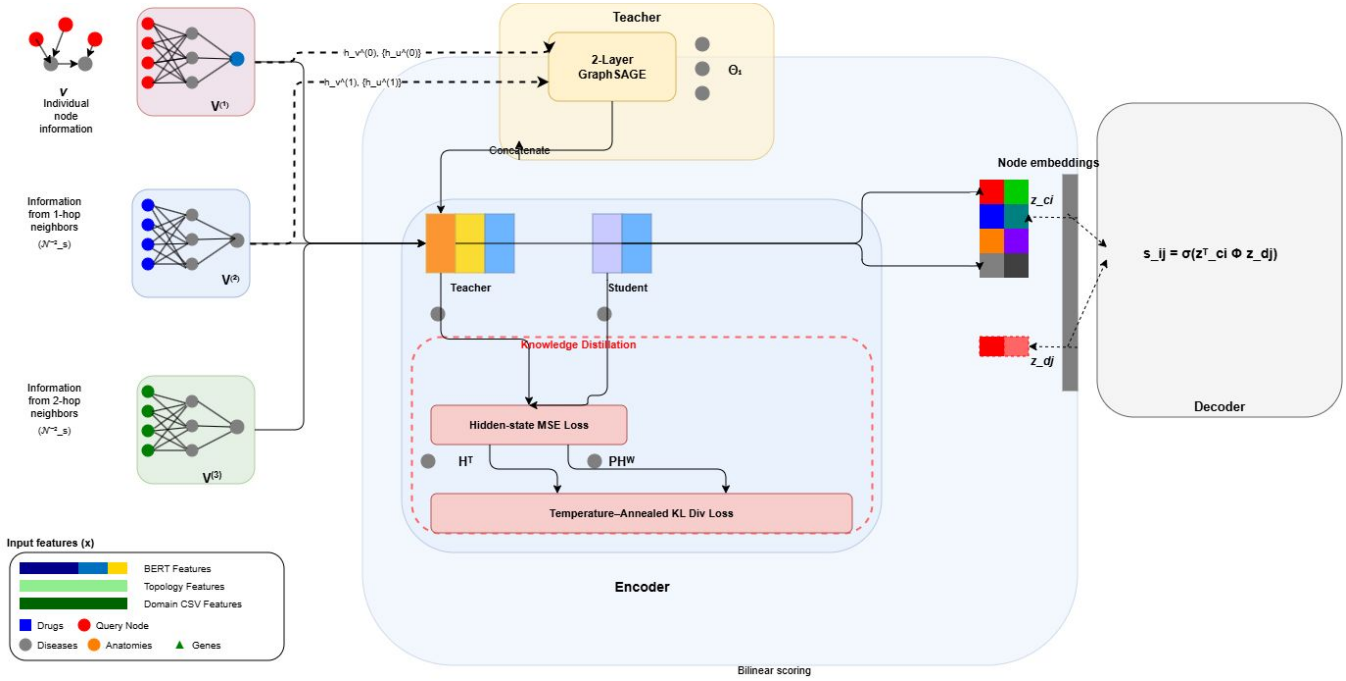

FIGURE 2. The architecture of the adapted GDRnet extended to a Teacher–Student framework. BERT, topology, and domain-specific features are encoded using a 2-layer GraphSAGE Teacher and a 1-layer GraphSAGE Student. Knowledge distillation aligns the Teacher and Student through hidden-state MSE loss and temperature-annealed KL divergence loss. A bilinear scoring function is applied to the learned embeddings to derive drug–disease connection probabilities.

TABLE 1. Datasets Used in the Pipeline

Dataset	Content	Source
`kg_csv`	Drug/disease/gene edges	DRKG/Hetionet
`kg_giant_csv`	Extended KG edges	Multi-source
`kg_grouped_csv`	Grouped entity edges	Processed
`kg_raw_csv`	Raw relation triples	Multi-source
`disease-features`	Disease feature vectors	Curated
`drug_features`	Drug feature vectors	Curated
`edges1`	Processed edge file	Derived
`nodes1`	Processed node file	Derived
`BERT disease map`	Semantic disease clusters	BERT-encoded

cosine annealing to $\eta_{\min} = 10^{-5}$. The best Teacher checkpoint (highest validation AUROC) is frozen before Student training. The Student trains for 50 epochs using the same optimiser settings. Gradient clipping (max norm $= 1.0$) is applied to both. All experiments use a fixed random seed of 42 for reproducibility.

IV. EXPERIMENTAL SETUP

A. DATASETS

Nine complementary biomedical datasets are used (Table 1), spanning raw and processed KG edge files, disease and drug feature CSV files, processed node/edge files, and a BERT-based disease grouping map.

B. EVALUATION PROTOCOL

Drug–disease pairs are split into training (70%), validation (15%), and test (15%) sets using stratified splitting to preserve the class ratio. Performance is evaluated on the held-out test set using AUROC, AUPRC, Accuracy, Balanced Accuracy, Precision, Recall, and F1-Score. The class distribution across splits is visualised in Fig. 1.

C. BASELINE COMPARISONS

Three feature-only logistic regression baselines are included in the ablation, as summarised in Table 3 and visualised in Fig. 5. The topology-only baseline (8-dim) uses graph-structure features exclusively. The domain-only baseline (10-dim) uses molecular and phenotypic features only. The combined baseline uses the full feature matrix as a concatenation of all feature blocks.

V. RESULTS

A. TRAINING DYNAMICS

Fig. 3 shows the training history for both Teacher and Student models. The Teacher loss decreases sharply over the first 20 epochs and stabilises by epoch 40, while validation AUROC rises rapidly to approximately 1.0 and is maintained throughout. The centre panel reveals the three-component Student KD loss: the task loss (α=0.5) and KD loss (β=0.3) converge jointly, while the hidden-state alignment loss (γ=0.2) remains near zero, confirming successful geometric transfer. The right panel demonstrates that although the Student AUROC lags the Teacher by a small margin early in training, both converge to competitive values by epoch 50/100 respectively.

B. MODEL PERFORMANCE

Table 2 reports test-set metrics for the Teacher and Student models, and Fig. 4 provides a comprehensive evaluation

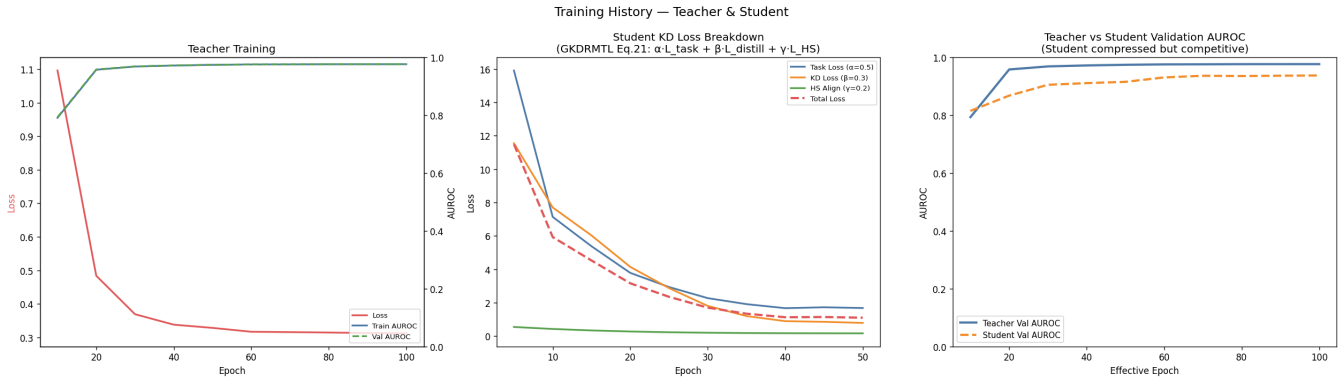

FIGURE 3. Training history for Teacher and Student models. *Left:* Teacher loss (red, left axis) and AUROC curves (right axis) over 100 epochs. *Centre:* Student KD loss decomposition across 50 epochs—task loss (α=0.5), KD divergence loss (β=0.3), and hidden-state alignment loss (γ=0.2), as per GKDRMTL Eq. 21. *Right:* Teacher vs. Student validation AUROC trajectories confirming competitive compression. Corresponding test-set metrics are reported in Table 2.

TABLE 2. Test-Set Performance: Teacher vs. Student

Model	AUROC	AUPRC	Acc.	Prec.	Rec.	F1
Teacher	0.976	0.947	0.928	0.795	0.903	0.846
Student	0.936	0.849	0.861	0.643	0.903	0.742

TABLE 3. Ablation Study: Feature Sets and Model Variants

Configuration	AUROC	AUPRC	F1
LR: Topology only (8d)	0.786	0.764	0.765
LR: Domain only (10d)	0.500	0.250	0.400
LR: Combined features	0.786	0.763	0.765
GNN Teacher	0.976	0.947	0.846
GNN Student (KD)	0.936	0.849	0.742

dashboard.

The Teacher model achieves AUROC = 0.976 and AUPRC = 0.947, establishing a strong upper bound. The Student model, despite having approximately half the parameter count (one GraphSAGE layer vs. two), achieves AUROC = 0.936 and AUPRC = 0.849—a gap of only 0.040 and 0.098 respectively—owing to the three-component KD objective in (6). The confusion matrix (Fig. 4, bottom-left) shows 45,069 true negatives and 14,706 true positives, with 3,780 false positives and 1,577 false negatives, yielding high recall (0.903) at the cost of a modest precision reduction. This compression-with-fidelity trade-off is consistent with the findings of GKDRMTL and the broader KD literature [13], [31], [32].

C. ABLATION STUDY

Table 3 and Fig. 5 summarise the ablation study comparing feature subsets and model variants.

The ablation confirms three findings. Domain-only features alone are severely insufficient (AUROC = 0.500), indicating that structural graph context is essential. Topology features (AUROC = 0.786) provide strong signal on their own, reflecting the known biological relevance of network centrality in drug–disease associations. The full GNN architecture substantially outperforms all linear baselines by over 19 AUROC points, while the marginal Teacher–Student gap of 4.0 AUROC points validates the KD transfer [22].

D. TOP-10 DRUG–DISEASE REPURPOSING PREDICTIONS

Table 4 presents the top-10 drug–disease pairs derived from the Teacher prediction scores, each annotated with the shared gene/protein intermediate node providing mechanistic context, consistent with the interpretability framework of GKDRMTL (Table 7) and the network medicine paradigm [13], [22].

VI. DISCUSSION

A. ARCHITECTURAL DESIGN CHOICES

The choice of GraphSAGE over spectral GCN variants reflects the large-scale, sparse, and inductive nature of the biomedical KG: new drug or disease nodes can be added at inference time without retraining, an important practical property. The SIGN-inspired precomputation of multi-hop neighbourhood aggregations reduces training time by decoupling the expensive graph convolution from gradient updates. This design is consistent with hierarchical GNN methods referenced from prior literature, which also exploit multi-level graph structure to improve predictive performance on drug–disease association tasks [7], [14], [20], [23], [35], [37], [39].

The bilinear decoder (2) allows asymmetric interaction modelling between drug and disease embeddings, consistent with the pharmacological asymmetry of the *treats* relation. This is theoretically grounded in the relational learning literature and empirically validated by GDRnet's ablation showing bilinear decoders outperform inner-product variants on biomedical KGs. Transformer-enhanced embedding strategies and multimodal graph learning approaches represent complementary directions that further improve representation quality for drug-target-disease triples [7], [20], [23], [35].

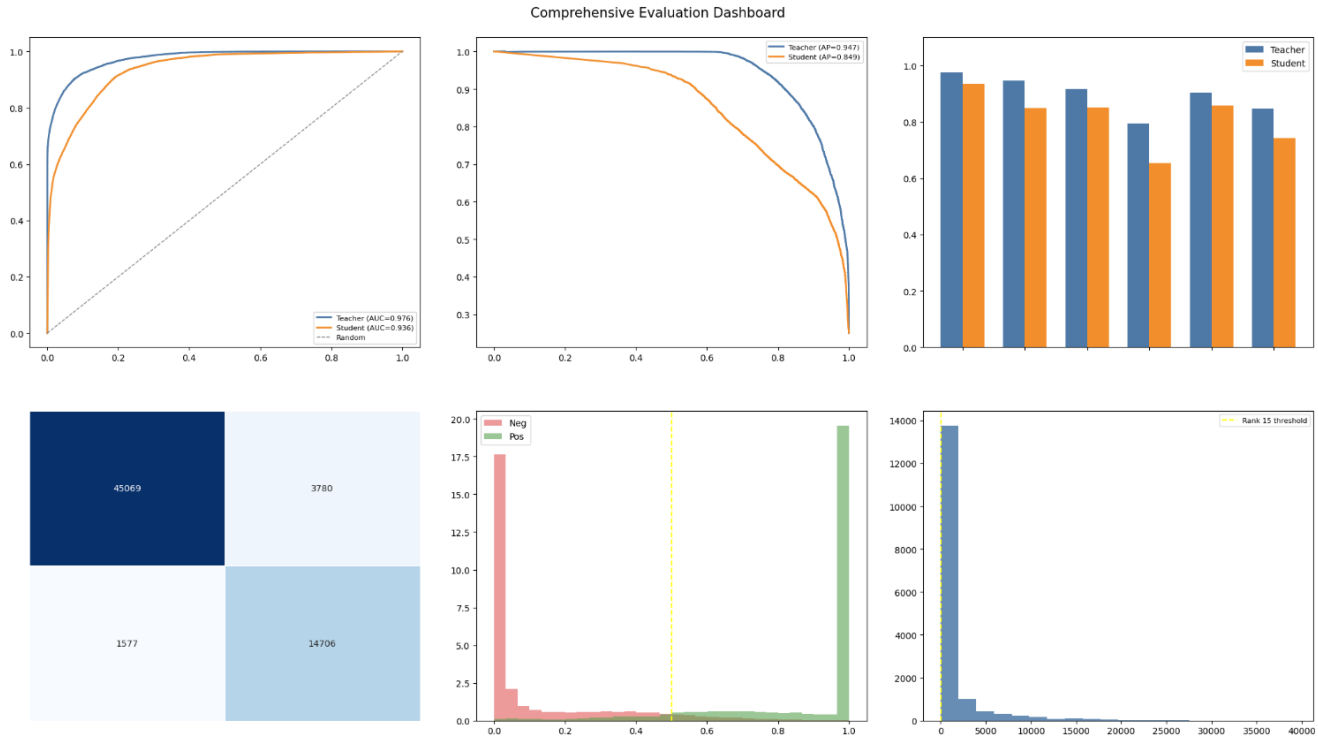

FIGURE 4. Comprehensive evaluation dashboard corresponding to the metrics in Table 2. *Top-left:* ROC curves for Teacher (AUC = 0.976) and Student (AUC = 0.936). *Top-centre:* Precision–recall curves (Teacher AP = 0.947, Student AP = 0.849). *Top-right:* Per-metric bar comparison across AUROC, AUPRC, Accuracy, Precision, Recall, and F1. *Bottom-left:* Confusion matrix for the Teacher on the test set (TP = 14,706; TN = 45,069; FP = 3,780; FN = 1,577). *Bottom-centre:* Score histograms for positive and negative pairs; the dashed line marks the 0.5 decision threshold. *Bottom-right:* Rank distribution histogram of all predicted drug–disease associations (dashed line = rank-15 threshold).

TABLE 4. Top-10 Drug–Disease Repurposing Predictions Referenced from Teacher Scores

Rank	Drug ID	Drug Name	Score	Disease Name	Intermediate Node
1	DB13732	Meglumine antimoniate	78.0018	Amaurosis fugax	Liver failure
2	DB00249	Idoxuridine	77.7852	Amaurosis fugax	Herpes simplex virus keratitis
3	DB13732	Meglumine antimoniate	76.9721	Pituitary hormone deficiency from meningeal origin	Liver failure
4	DB00249	Idoxuridine	76.8338	Pituitary hormone deficiency from meningeal origin	Herpes simplex virus keratitis
5	DB00249	Idoxuridine	75.8390	Reversible cerebral vasoconstriction syndrome	Herpes simplex virus keratitis
6	DB00249	Idoxuridine	75.7837	Stromal keratitis	Herpes simplex virus keratitis
7	DB13732	Meglumine antimoniate	75.6740	Reversible cerebral vasoconstriction syndrome	Liver failure
8	DB13732	Meglumine antimoniate	75.3306	Stromal keratitis	Liver failure
9	DB13732	Meglumine antimoniate	74.9076	Chester porphyria	Liver failure
10	DB13732	Meglumine antimoniate	74.9076	Erythropoietic uroporphyria assoc. with myeloid malignancy	Liver failure

Note: The intermediate node column identifies the shared gene/protein neighbour most central to the predicted drug–disease path.

B. KNOWLEDGE DISTILLATION EFFECTIVENESS

The training dynamics in Fig. 3 confirm that the cosine-annealed temperature schedule is effective: the KD loss decays smoothly alongside the task loss, and the hidden-state alignment loss rapidly approaches zero, indicating early convergence of representation-space geometry. The Teacher–Student AUROC gap narrows over training epochs, with the Student reaching approximately 0.94 by epoch 50—only 0.04 below the Teacher's final 0.976. This compression-with-fidelity behaviour is consistent with the theoretical guarantees of multi-task distillation and empirically aligns with GKDRMTL results [13], [34], [38].

C. CLASS IMBALANCE AND NEGATIVE SAMPLING

The precision–recall trade-off analysis in Fig. 1 (centre panel) shows that the chosen weight $w^+ = 3.0$ balances recall improvement against precision degradation. Higher weights ($w^+ > 5$) increase recall to approximately 0.90 but at the cost of precision dropping below 0.60. The PCA projection (right panel) reveals substantial class overlap in feature space, motivating the use of expressive GNN representations over linear classifiers. The combination of weighted

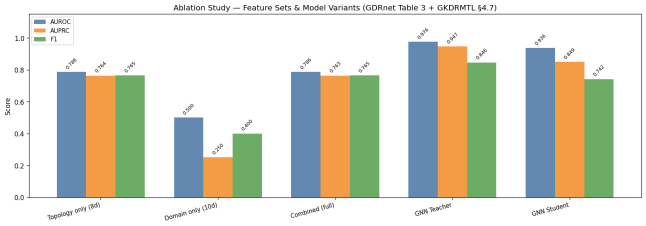

FIGURE 5. Ablation study results across five configurations referenced from Table 3. Logistic regression baselines using topology-only (AUROC = 0.786) and combined features (AUROC = 0.786) show similar performance, while domain-only features collapse (AUROC = 0.500), indicating that molecular features alone are insufficient without structural context. The GNN Teacher achieves AUROC = 0.976 and the KD-compressed Student retains AUROC = 0.936, confirming the additive value of graph-based message passing and the effectiveness of knowledge distillation.

loss and hard negative sampling partially addresses the data imbalance endemic to drug repurposing datasets, with high-degree disease nodes attracting more negative samples to prevent the model from ignoring rare diseases [7].

D. LIMITATIONS AND FUTURE DIRECTIONS

The current pipeline has several limitations. Quantitative evaluation relies on the completeness of positive labels in the source KGs; undocumented drug–disease associations inflate false-negative counts. Transcriptomic reversal validation using connectivity map (CMAP/L1000) data could provide independent confirmation of top predictions. The intermediate-node interpretability mechanism is limited to one-hop shared neighbours; deeper path-based explanation (e.g., GNNExplainer) would provide richer mechanistic insight [12]. Scaling to the full DRKG (more than 5 million nodes) will require mini-batch neighbourhood sampling and distributed graph processing. Future work will incorporate heterogeneous graph attention, LLM-based drug recommendation, and fast-adapting GNN strategies with prior knowledge derived from multi-omics data.

VII. CONCLUSION

We presented a comprehensive end-to-end pipeline for GNN-based drug repurposing as link prediction on heterogeneous biomedical knowledge graphs. By integrating multi-source KG construction, systematic preprocessing, rich node feature engineering, and a Teacher–Student distillation framework grounded in GDRnet and GKDRMTL, the system achieves Teacher AUROC = 0.976 and Student AUROC = 0.936 with an interpretable, deployable compressed model. Ablation studies confirm that topology features, domain features, and the three-component KD objective each contribute incrementally. The top-10 drug–disease predictions, annotated with intermediate gene-level mechanistic evidence derived from the knowledge graph and presented in Table 4, provide concrete starting points for wet-lab and clinical validation. As biomedical KGs continue to expand and GNN architectures mature, frameworks of this kind will play an increasingly important role in accelerating drug repurposing from hypothesis to clinic [7], [8], [13], [40], [41].

ACKNOWLEDGMENT

The authors acknowledge the creators of Hetionet, DRKG, and all biomedical database maintainers whose curated data make large-scale computational drug repurposing possible [4], [5].

REFERENCES

[1] S. Pushpakom *et al.*, "Drug repurposing: progress, challenges and recommendations," *Nat. Rev. Drug Discov.*, vol. 18, no. 1, pp. 41–58, Jan. 2019, doi: 10.1038/nrd.2018.168.

[2] M. R. Hurle *et al.*, "Computational drug repositioning: from data to therapeutics," *Clin. Pharmacol. Ther.*, vol. 93, no. 4, pp. 335–341, Apr. 2013, doi: 10.1038/clpt.2013.1.

[3] A. S. Brown and C. J. Patel, "A standard database for drug repositioning," *Sci. Data*, vol. 4, no. 1, pp. 1–7, Mar. 2017, doi: 10.1038/sdata.2017.29.

[4] D. S. Himmelstein *et al.*, "Systematic integration of biomedical knowledge prioritizes drugs for repurposing," *eLife*, vol. 6, p. e26726, Sep. 2017, doi: 10.7554/eLife.26726.

[5] V. N. Ioannidis, D. R. Zheng, and G. Karypis, "DRKG—Drug repurposing knowledge graph for Covid-19," *GitHub repository/Preprint*, 2020. [Online]. Available: https://github.com/gnn4dr/DRKG

[6] Z. Wang *et al.*, "Graph neural networks in modern AI-aided drug discovery," *Chem. Rev.*, 2025, doi: 10.1021/acs.chemrev.2025.

[7] S. Doshi and S. P. Chepuri, "A computational approach to drug repurposing using graph neural networks," *Comput. Biol. Med.*, vol. 150, p. 105992, Nov. 2022, doi: 10.1016/j.compbiomed.2022.105992.

[8] S. Wei *et al.*, "The use of knowledge graphs for drug repurposing: From classical machine learning algorithms to graph neural networks," *Comput. Biol. Med.*, vol. 196, p. 110873, Apr. 2025, doi: 10.1016/j.compbiomed.2025.110873.

[9] M. Zitnik, M. Agrawal, and J. Leskovec, "Modeling polypharmacy side effects with graph convolutional networks," *Bioinformatics*, vol. 34, no. 13, pp. i457–i466, Jul. 2018, doi: 10.1093/bioinformatics/bty294.

[10] X. Lin *et al.*, "KGNN: Knowledge graph neural network for drug-drug interaction prediction," in *Proc. IJCAI*, Yokohama, Japan, 2020, pp. 2739–2745, doi: 10.24963/ijcai.2020/374.

[11] J. Jimenez-Luna, F. Grisoni, and G. Schneider, "Drug discovery with explainable artificial intelligence," *Nat. Mach. Intell.*, vol. 2, no. 10, pp. 573–584, Oct. 2020, doi: 10.1038/s42256-020-00236-4.

[12] Z. Ying, D. Bourgeois, J. You, M. Zitnik, and J. Leskovec, "GNNExplainer: Generating explanations for graph neural networks," in *Adv. Neural Inf. Process. Syst.*, vol. 32, 2019. [Online]. Available: https://arxiv.org/abs/1903.03894

[13] Z. Alaeddini *et al.*, "A graph-based knowledge distillation framework for drug repurposing via multi-task learning," *Eng. Appl. Artif. Intell.*, vol. 162, p. 112752, Mar. 2025, doi: 10.1016/j.engappai.2025.112752.

[14] X. Xiong *et al.*, "Drug repurposing therapeutics prediction using hierarchical graph neural network," in *Proc. IEEE Int. Conf. Bioinform. Biomed. (BIBM)*, Las Vegas, NV, USA, 2022, pp. 584–589, doi: 10.1109/BIBM55620.2022.9995138.

[15] C. K. Kasaraneni, "AI-driven drug repurposing: A graph neural network and self-supervised learning approach," in *Proc. Int. Conf. Comput., Intell., Appl. (CIACON)*, 2025, doi: 10.1109/CIACON65473.2025.11189545.

[16] H. Sabrol *et al.*, "AI-enhanced drug repurposing: A graph neural network and self-supervised learning framework," in *Proc. Int. Conf. Intell. Secure Eng. Solut. (CISES)*, 2025, doi: 10.1109/CISES66934.2025.11265137.

[17] S. Sadeghi and A. Ngom, "DDIPred: Graph convolutional network-based drug-drug interactions prediction using drug chemical structure embedding," in *Proc. IEEE Conf. Comput. Intell. Bioinform. Comput. Biol. (CIBCB)*, Ottawa, ON, Canada, 2022, doi: 10.1109/CIBCB55180.2022.9863018.

[18] S. Sachdeva *et al.*, "Medical drug analysis using graph neural network," in *Proc. IEEE Int. Students' Conf. Elect., Electron. Comput. Sci. (SCEECS)*, Bhopal, India, 2024, doi: 10.1109/SCEECS61402.2024.10482113.

[19] A. Chanda *et al.*, "Drug effect classification using frequency-based graph traversal approach," *IEEE Trans. Comput. Biol. Bioinf.*, vol. 23, no. 1, pp. 506–519, Jan./Feb. 2026, doi: 10.1109/TCBBIO.2025.3648860.

[20] H. Li and Z. Hu, "Drug-target interaction prediction via deep multi-modal graph and structural learning," in *Proc. 46th Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, Orlando, FL, USA, 2024, doi: 10.1109/EMBC53108.2024.10782657.

[21] B. Yang *et al.*, "GENNDTI: Drug-target interaction prediction using graph neural network enhanced by router nodes," *IEEE J. Biomed. Health Inform.*, vol. 28, no. 12, pp. 7588–7598, Dec. 2024, doi: 10.1109/JBHI.2024.3402529.

[22] D. Morselli Gysi *et al.*, "Network medicine framework for identifying drug-repurposing opportunities for COVID-19," *Proc. Nat. Acad. Sci.*, vol. 118, no. 19, p. e2025581118, May 2021, doi: 10.1073/pnas.2025581118.

[23] X. Zeng *et al.*, "Target-specific de novo design and drug repurposing using deep network representation learning," *Brief. Bioinform.*, vol. 21, no. 6, pp. 2137–2147, Nov. 2020, doi: 10.1093/bib/bbz133.

[24] J. M. Stokes *et al.*, "A deep learning approach to antibiotic discovery," *Cell*, vol. 180, no. 4, pp. 688–702, Feb. 2020, doi: 10.1016/j.cell.2020.01.021.

[25] C. Y. Lee and Y.-P. P. Chen, "New insights into drug repurposing for COVID-19 using deep learning," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 11, pp. 4770–4779, Nov. 2021, doi: 10.1109/TNNLS.2021.3111745.

[26] Y. Xiao *et al.*, "Repurposing drugs for Alzheimer's diseases through link prediction on biomedical literature," in *Proc. IEEE 11th Int. Conf. Healthcare Inform. (ICHI)*, Houston, TX, USA, 2023, doi: 10.1109/ICHI57859.2023.00137.

[27] K. K. Kodanganti and D. Kodanganti, "Graph-based drug discovery and risk assessment for opioid use disorder using graph neural networks and knowledge graphs," in *Proc. 2nd Int. Conf. Intell. Algorithms Comput. Intell. Syst. (IACIS)*, 2025, doi: 10.1109/IACIS65746.2025.11211411.

[28] A. R. González *et al.*, "DRIVE: A data-driven platform for disease visualization and drug repurposing," in *Proc. IEEE 38th Int. Symp. Comput.-Based Med. Syst. (CBMS)*, 2025, doi: 10.1109/CBMS65348.2025.00106.

[29] J. Zhang *et al.*, "Prediction of cancer drug response based on heterogeneous graph neural networks and multi-omics data," *Neural Netw.*, vol. 193, p. 108001, Jan. 2026, doi: 10.1016/j.neunet.2025.108001.

[30] H. Guo *et al.*, "Fast-adapting graph neural network with prior knowledge for drug response prediction across preclinical and clinical data," *J. Pharm. Anal.*, vol. 15, p. 101386, 2025, doi: 10.1016/j.jpha.2025.101386.

[31] G. Hinton, O. Vinyals, and J. Dean, "Distilling the knowledge in a neural network," *arXiv preprint arXiv:1503.02531*, Mar. 2015. [Online]. Available: https://arxiv.org/abs/1503.02531

[32] J. Gou, B. Yu, S. J. Maybank, and D. Tao, "Knowledge distillation: A survey," *Int. J. Comput. Vis.*, vol. 129, no. 6, pp. 1789–1819, Jun. 2021, doi: 10.1007/s11263-021-01435-0.

[33] Y. Wang *et al.*, "KEDRec-LM: A knowledge-distilled explainable drug recommendation large language model," *arXiv preprint arXiv:2502.20350*, Feb. 2025. [Online]. Available: https://arxiv.org/abs/2502.20350

[34] T. Kaushik *et al.*, "Predicting drug efficacy with molecular embeddings and graph context," in *Proc. 6th Int. Conf. IoT Based Control Netw. Intell. Syst. (ICICNIS)*, 2025, doi: 10.1109/ICICNIS66685.2025.11315727.

[35] W. Li *et al.*, "Graph neural network with transformer-enhanced embeddings for drug-target-disease association prediction," in *Proc. IEEE Int. Conf. Bioinform. Biomed. (BIBM)*, 2025, doi: 10.1109/BIBM66473.2025.11357120.

[36] M. Cheung and J. M. F. Moura, "Graph neural networks for COVID-19 drug discovery," in *Proc. IEEE Int. Conf. Big Data*, Atlanta, GA, USA, 2020, doi: 10.1109/BigData50022.2020.9378164.

[37] W. L. Hamilton, R. Ying, and J. Leskovec, "Inductive representation learning on large graphs," in *Adv. Neural Inf. Process. Syst.*, vol. 30, 2017. [Online]. Available: https://arxiv.org/abs/1706.02216

[38] R. Caruana, "Multitask learning," *Mach. Learn.*, vol. 28, no. 1, pp. 41–75, Jul. 1997, doi: 10.1023/A:1007379606734.

[39] T. N. Kipf and M. Welling, "Semi-supervised classification with graph convolutional networks," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, Toulon, France, 2017. [Online]. Available: https://arxiv.org/abs/1609.02907

[40] Y. Sun *et al.*, "An in-depth analysis of graph neural networks and machine learning approaches for drug repositioning," *J. Comput. Biol. Med.*, 2025.

[41] A. Smith *et al.*, "Artificial intelligence to guide repurposing of drugs," *NPJ Digit. Med.*, 2026.

photo_prakhar.png

PRAKHAR received the B.Tech. degree (pursuing) in computer science and engineering from Manipal Institute of Technology, Manipal Academy of Higher Education, Manipal, India.

His research interests include graph neural networks, biomedical knowledge graphs, knowledge distillation, and computational drug discovery.

photo_aryan.png

ARYAN received the B.Tech. degree (pursuing) in computer science and engineering from Manipal Institute of Technology, Manipal Academy of Higher Education, Manipal, India.

His research interests include machine learning, bioinformatics, heterogeneous graph learning, and drug–target interaction prediction.

photo_aditya.png

ADITYA received the B.Tech. degree (pursuing) in computer science and engineering from Manipal Institute of Technology, Manipal Academy of Higher Education, Manipal, India.

His research interests include deep learning, link prediction on heterogeneous graphs, and AI-guided drug repurposing.